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Quantum Computing Applications in Financial Modeling and Portfolio Optimization

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Abstract
In recent years, advancements in classical computing have grown rapidly; however, quantum computing is expected to surpass traditional computational limits and revolutionize multiple industries. Among these, the financial sector is anticipated to be one of the earliest beneficiaries, particularly for medium- and long-term analytical tasks. This study investigates the role of quantum computation in financial modeling and portfolio optimization, emphasizing the value of classical convex optimization algorithms integrated with emerging quantum methods. Key areas in which quantum computing demonstrates potential include machine learning–based optimization, stochastic analysis, fraud detection, natural language processing, portfolio construction, risk estimation, and derivative valuation. Moreover, the practicality of deploying such algorithms on near-term quantum hardware is explored, along with the suitability of various architectures for diverse financial use cases.

Keywords: Quantum Computing, Financial Modeling, Portfolio Optimization, Quantum Algorithms

Introduction

Quantum computation refers to the storage and manipulation of information using principles that differ fundamentally from classical computing. Unlike classical systems that operate based on deterministic classical mechanics, quantum computers rely on the rules of quantum mechanics—particularly superposition, entanglement, and probabilistic state behavior. These quantum effects typically appear only at extremely small scales and require delicate isolation from environmental interference, making the construction of stable quantum systems highly challenging.

Interest in quantum technologies has surged across physics, computer science, and applied fields. Finance, in particular, stands out as an area where complex, computationally intensive problems are abundant, making it well-aligned with the strengths of quantum algorithms. Various financial applications—including risk modeling, portfolio optimization, and asset pricing—present opportunities for significant improvement through quantum methods. However, navigating the extensive and fragmented research landscape poses difficulties for practitioners seeking a unified understanding of quantum-enabled financial systems.

Quantum mechanics, the branch of physics that studies matter and energy at atomic and subatomic scales, provides the theoretical foundation for quantum computation. Developed in the early twentieth century, it describes physical behavior in probabilistic terms rather than deterministic ones. Quantum principles such as superposition and entanglement are central to the power of quantum algorithms.

Portfolio management, a cornerstone of financial decision-making, involves the optimal selection of assets—such as equities, bonds, commodities, and loans—to meet specific investment objectives. The classical foundation for portfolio optimization stems from Harry Markowitz's Modern Portfolio Theory (MPT), which highlights the importance of balancing expected returns with associated risk. Since its introduction, various adaptations of the theory have emerged, incorporating multiple objectives, stochastic and deterministic variables, transaction costs, and dynamic time horizons.

As financial institutions increasingly rely on precise statistical models capable of capturing a wide range of potential outcomes, the demand for more powerful computational tools has grown. Quantum computing, with its ability to process large datasets and handle complex optimization spaces, offers promising improvements for numerous financial challenges. This study explores major financial use cases poised to benefit from quantum algorithms, while emphasizing their potential contribution to risk management and decision-making in modern markets.

Methodology

Portfolio optimization involves strategically managing a set of assets to maximize returns while minimizing risk. The methodology used in this study outlines essential components required to formulate and solve such optimization problems. Key inputs include asset characteristics—expected returns, risks, and correlations—followed by the specification of constraints and objective functions. A real-world case study involving outstanding bank loans is used to illustrate the methodology, incorporating considerations such as carbon-reduction targets, concentration risks, and return-on-capital expectations.

Review of Literature

A. Quantum Computing in Finance

Quantum computing continues to evoke both enthusiasm and skepticism. While quantum devices have demonstrated the ability to outperform classical systems in specialized tasks, numerous applications remain unexplored or require further refinement before achieving general utility. This research focuses on the application of quantum optimization for portfolio construction. Empirical analyses reveal how parameter choices—such as chain strength, number of reads, annealing schedules, and embedding strategies—affect solution quality in quantum annealers.

The integration of quantum techniques with machine learning represents another promising avenue. Quantum-enhanced machine learning algorithms show potential for improving portfolio selection, enabling higher returns and more efficient risk management. This study highlights advances in quantum support vector machines (QSVM) and discusses challenges such as circuit optimization, noise mitigation, and hybrid quantum-classical integration.

B. Portfolio Optimization Model

Let a portfolio consist of N assets $P_1, P_2, \ldots, P_n$, each defined by its expected return μ_i and risk σ_i. Asset relationships are captured through correlation coefficients ρ_{ij}. A binary variable x_i indicates whether asset i is included in the portfolio. The optimization problem seeks to minimize total risk subject to constraints on expected return and asset count.

The risk matrix Σ is defined as:

$\sigma_{ij} = \sigma^2_i$ if $i = j$

$\sigma_{ij} = \rho_{ij}\, \sigma_i\, \sigma_j$ if $i \neq j$

The objective becomes:
minimize $\Sigma \Sigma x_i x_j \sigma_{ij}$
subject to:
$\Sigma x_i = N$ and $\mu^T x \geq R^*$

C. Case Study Formulation

A real-world case examining a bank's loan distribution strategy is presented. The goal is to allocate credit across geographic regions and sectors by 2030 while meeting sustainability metrics aligned with the Paris Agreement. Emission intensity, sector diversification (HHI constraints), and return-on-capital requirements are incorporated into the optimization problem.

Emission intensity E is measured as:
$E = \Sigma(e_i y_i) / \Sigma(y_i)$
Constraints require:
$0.76 \Sigma x_i e_i \leq 0.7E \Sigma x_i$
The complete optimization problem is:
minimize ϕ HHI(x) – (1 – ϕ) ROC(x)
subject to:
$LBi \leq x_i \leq Ubi$
and emission, ROC, and diversification constraints.

D. QUBO Formulation

Quantum annealers require Quadratic Unconstrained Binary Optimization (QUBO) expressions. The reformulated QUBO becomes:
$QUBO = \lambda_1 \Sigma x_i^2 - \lambda_2 \Sigma \theta_i x_i + \lambda_3 (0.76 \Sigma x_i e_i - 0.7E \Sigma x_i)$
Convex Optimization

Traditional convex optimization techniques remain foundational for solving multi-objective portfolio problems. Using linear matrix inequalities (LMIs), the problem can be solved using MATLAB-YALMIP with the SeDuMi solver. The resulting Pareto frontier is presented, offering insight into trade-offs between competing objectives.

Results and Discussion

Random sampling and simulated annealing were used to evaluate the QUBO-based portfolio model. A brute-force sampling approach generated over 250,000 solutions, though few satisfied emission constraints. Comparisons reveal that simulated annealing produces solutions closer to the Pareto frontier than quantum annealing in this experiment. The findings suggest that parameter tuning and hardware-specific optimization may significantly improve quantum performance in future applications. The broader set of solutions produced by the proposed approach offers alternative pathways and deeper insights beyond the classical optimal set.

Conclusion

This study demonstrates the potential of quantum computing for tackling complex portfolio optimization problems, particularly those involving multi-objective criteria and non-linear constraints. While classical quadratic programming approaches face limitations, quantum methods—especially quantum and simulated annealing—offer promising alternatives for navigating high-dimensional solution spaces. Quantum computing also holds potential across various financial tasks including fraud detection, natural

language processing, derivative pricing, and stochastic modeling. The QUBO framework provides a structured means to encode optimization challenges suited for quantum devices. Future efforts should focus on refining quantum algorithms, enhancing hardware capabilities, and improving hybrid quantum-classical workflows to unlock the full benefits of quantum technologies in finance.